

Using waste poly(vinyl chloride) to synthesize chloroarenes by plasticizer-mediated electro(de)chlorination

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New approaches are needed to both reduce and reuse plastic waste. In this context, poly(vinyl chloride) (PVC) is an appealing target as it is the least recycled high-production-volume polymer due to its facile release of plasticizers and corrosive HCl gas. Herein, these limitations become advantageous in a paired-electrolysis reaction in which HCl is intentionally generated from PVC to chlorinate arenes in an air- and moisture-tolerant process that is mediated by the plasticizer. The reaction proceeds efficiently with other plastic waste present and a commercial plasticized PVC product (laboratory tubing) can be used directly. A simplified life-cycle assessment reveals that using PVC waste as the chlorine source in the paired-electrolysis reaction has a lower global warming potential than HCl. Overall, this method should inspire other strategies for repurposing waste PVC and related polymers using electrosynthetic reactions, including those that take advantage of existing polymer additives.

Plastics recycling has until now been predominantly focused on mechanical recycling, which downgrades material quality and therefore limits its utility. Recycling rates (<10%)¹ are unlikely to improve with mechanical recycling alone, leading to calls for innovations in chemical recycling^{2,3}. Closed-loop chemical recycling—in which polymers are first depolymerized and then repolymerized to generate pristine material—has only been realized for hydrolysable polymers such as poly(ethylene terephthalate) (PET), whereas polymers with all-carbon backbones are not thermodynamically amenable⁴. The latter could be more practically recycled in an open-loop process, wherein polymers are broken down into smaller fragments and/or transformed into different materials that serve as feedstocks in other chemical processes. The leading open-loop process (high-temperature anaerobic pyrolysis) is only useful for hydrocarbon polymers such as polyethylene, polypropylene and polystyrene. Meanwhile, poly(vinyl chloride) (PVC) is not recycled through any of these approaches and has the lowest recycling rate in most countries (0% in the US), even though it has a high production rate⁵. Achieving a circular economy for plastics will

require recycling approaches that address each component of the complex waste stream^{6,7}.

As PVC poses a known health risk in the environment (for example, PVC has been detected in wastewater sludge and in aquatic life commonly consumed by humans), these low recycling rates are especially concerning^{8,9}. Poly(vinyl chloride) is ranked as the most hazardous polymer¹⁰, in part due to the plasticizing additives such as phthalates¹¹—which are carcinogenic and mutagenic—and corrosive chlorine-containing by-products that are formed upon degradation. These additives and by-products are also the reason why it is difficult to recycle PVC via mechanical approaches; plasticizing additives (often 10–70% by mass) mean that PVC must be separated from other plastics to avoid cross-contamination. Because complete separation is challenging, one of the most common PVC plasticizers (di(2-ethylhexyl) phthalate, DEHP) has been found in other recycled plastics¹². Even without additives, PVC is still not recycled by melt-processing or pyrolysis because hydrochloric acid (HCl) and other volatiles are rapidly eliminated upon heat treatment, the former of which poses a safety hazard

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and corrodes equipment¹³. From a waste-management perspective, removing both DEHP and chlorine from PVC waste would enable safer disposal of the remaining material by landfilling, mechanical recycling or pyrolysis.

Controlled PVC dechlorination relies on either thermal activation¹⁴ or stoichiometric strong oxidants¹⁵. As an example, hydrothermal treatments degrade DEHP and dechlorinate PVC at temperatures of over 250 °C in pressurized systems with catalysts or other additives^{16–18}. Although these treatments are useful for recovering the hydrocarbon content of PVC, the Cl content (57% by mass) is wasted. Although rapid HCl elimination from PVC is undesirable in mechanical recycling, HCl is a useful reagent in many chemical reactions¹⁹. When reimagined from a synthetic perspective, PVC waste could instead be viewed as an easy-to-handle solid reagent that releases HCl on demand. In situ-generated HCl (or Cl₂) could find a range of synthetic applications; for example, more than 85% of pharmaceuticals employ chlorine chemistry during production²⁰. In 2020, over 9,000 kt of chlorine was manufactured in Europe, with ~33% used in PVC and ~54% used in other organochlorine applications²¹. With the method described herein, the fraction of chlorine stored in PVC could later be used in synthetic reactions.

Herein we explore whether HCl can be generated on demand from waste PVC and repurposed to chlorinate organic molecules. More specifically, we propose that chloride anions generated from PVC under electroreductive conditions could be directly used in a tandem electrooxidative chlorination reaction (Fig. 1a). Such paired-electrolysis reactions (strategic combinations of reduction and oxidation half-reactions) are often used to generate reactive halogen species (reductively at the cathode) and then halogenate a substrate (oxidatively at the anode)^{22–25}. One advantage of this approach is that the plasticizer, DEHP²⁶—a problem when it comes to recycling—could serve as a redox mediator. Another advantage is the direct use of electrons, which is inexpensive, non-toxic and has a small ecological footprint^{27–29}. A simplified life-cycle assessment (LCA) of only the materials and end-of-life stages determines the global warming potential (GWP) of the electrochlorination using PVC versus pristine HCl. Overall, this work represents an approach to handling waste PVC that takes advantage of its high degree of functional groups in an electro(de)chlorination mediated by its plasticizer to chlorinate organic molecules.

Results and discussion

Inspiration from the literature

The electroreduction of PVC was previously reported by a single group of scientists, although these findings have become obscured in the literature^{30–32}. In their brief 1987 communication, Shapoval and co-workers noted that solution-phase PVC eliminated HCl on exposure to negative voltage (–2.2 V versus the saturated calomel electrode). The same reactivity occurred at a smaller potential with the plasticizer dioctyl phthalate present (–1.8 V versus the saturated calomel electrode). Although their work demonstrated that PVC is susceptible to electroreductive dehydrochlorination and a common plastic additive may facilitate this reactivity, the polymer product was never characterized and no further studies on this system have been reported. More recently, Jiao and co-workers²³ reported a paired-electrolysis reaction using the small molecule dichloroethane as the chloride source; however, no redox mediators were used to improve the reactivity or yield in this process, and expensive platinum cathodes were used. Based on these precedents, however, we began evaluating the paired-electrolysis reaction using PVC as the chloride source and DEHP as the redox mediator to chlorinate organic molecules.

Paired-electrolysis using PVC as a chloride source

The paired-electrolysis was performed in an undivided cell using ethoxybenzene (**1**) as the substrate (Fig. 1b). Dimethylformamide (DMF) was selected as the solvent as it has a wide electrochemical window and

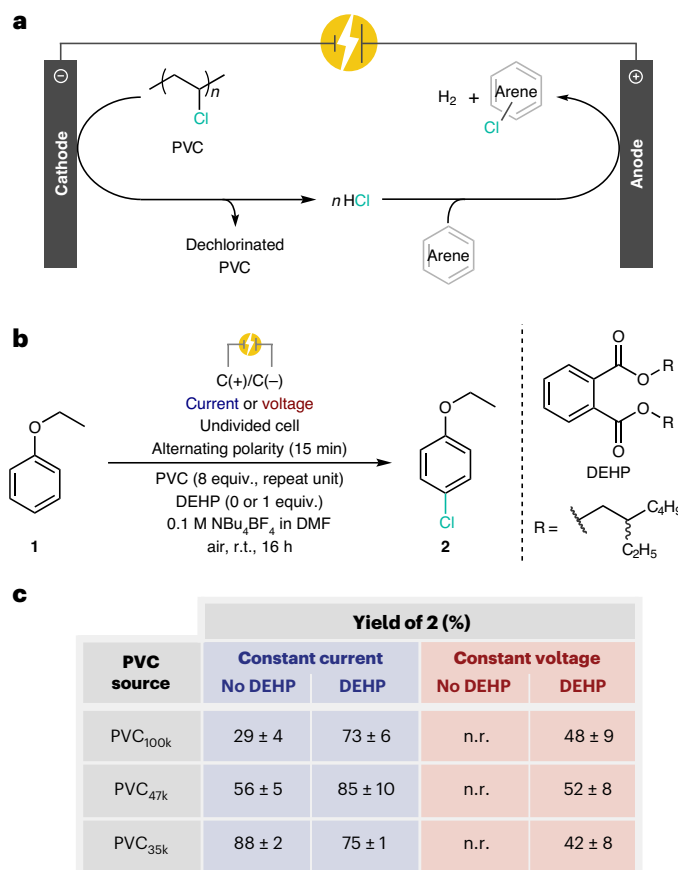


Fig. 1 | Paired-electrolysis reaction. a, Scheme for reductive dechlorination of PVC (cathode) and oxidative chlorination of an arene (anode). **b**, The model reaction and conditions evaluated in this work. Ethoxybenzene (**1**) is chlorinated using PVC and DEHP. **c**, Summary of results showing that the chlorination reaction proceeded to higher yields with the lower-molar-mass PVC, and that reactivity can be turned on/off with the plasticizer present/absent under constant-voltage conditions. Note that the reported yields are an average of two runs ± s.d., and were measured by GC-MS. n.r., no reaction.

dissolves (or swells) all reaction components. Graphite was selected for both electrodes because it is relatively inexpensive, and having identical electrodes enabled us to alternate the polarity to minimize polymer build-up on the electrodes. Tetrabutylammonium tetrafluoroborate (NBu₄BF₄) was chosen as the electrolyte so that the non-nucleophilic anion would not interfere in the synthetic reactions. Three different number-average molar masses (*M_n*) of PVC were used: PVC_{35k}, PVC_{47k} and PVC_{100k}. The two lower-molar-mass polymers completely dissolved in DMF at room temperature, whereas PVC_{100k} only partially dissolved and swelled. Reaction screening was first performed without DEHP present. Under constant-current electrolysis (galvanostatic conditions at 10 mA), the chlorination reaction proceeded to higher yields with the lower-molar-mass PVC (88% for PVC_{35k}), even though the total mass of polymer added was consistent in each case (Fig. 1c). This molar mass dependence may be due to the increased diffusivity of smaller macromolecules compared with larger ones. Conformational differences may also play a role, wherein the larger macromolecules are more coiled³³ and therefore have less accessible C–Cl bonds³⁴. Furthermore, at the same mass loading, the lower-molar-mass polymers have more individual chains leading to more frequent collisions with the electrode. Nevertheless, with all PVC sources, build-up of insoluble black residue was observed on the electrodes after the reaction, suggesting that PVC was being reduced at the electrode surface.

Paired-electrolysis using both PVC and plasticizer

Next, the same constant-current paired-electrolyses were performed with DEHP present: the reactions with PVC_{47k} and PVC_{100k} proceeded to higher yields (85% and 73%, respectively), whereas the already high yield using PVC_{35k} decreased slightly to 75%. Little-to-no insoluble polymer residue was observed on the electrodes, suggesting that PVC was being reduced indirectly rather than directly at the electrode surface.

To further probe the impact of DEHP, the same reactions were performed under constant-voltage conditions (potentiostatic conditions at -1.3 V versus Ag/AgCl), which is the smallest voltage that showed reactivity using an IKA ElectraSyn. When DEHP was included, the chlorination reaction proceeded but did not depend on M_n (42–52% yield). By contrast, no reaction was observed without DEHP (0% yield). The on/off switch in reactivity, with and without DEHP, indicates that PVC alone is unreactive at the set voltage, and that DEHP plays a role in the paired-electrolysis reaction, possibly as a redox mediator. This on/off reactivity was not observed under galvanostatic (constant-current) conditions because the working potentials of the electrode automatically adjusted to the larger potentials needed to reduce PVC directly. Given these results, the role of DEHP was further evaluated using electroanalytical experiments³⁵.

PVC attenuates reoxidation of DEHP radical anion

A cyclic voltammogram of PVC alone revealed that it does not exhibit any redox activity within the solvent window. By contrast, DEHP undergoes two successive reductions ($E_{1/2} = -2.56$ V and -2.74 V versus Ag/Ag⁺) with a redox profile similar to one previously reported in acetonitrile (Supplementary Fig. 12)³⁶. The first reduction is semi-reversible ($i_{pa}/i_{pc} < 1$), whereas the second reduction is essentially irreversible ($i_{pa}/i_{pc} = 0$), even at a fast scan rate (100 mV s⁻¹). The reversibility of the first DEHP reduction decreases when PVC is added, which suggests electron transfer is occurring between the two species. Using a protocol similar to the mediated electrochemical reduction of butyl halides³⁷, we added increasingly higher concentrations of PVC_{35k}, PVC_{47k} or PVC_{100k} ([PVC] = 0–30 mM in repeat units) to a solution of DEHP (1 mM) and measured the reversibility of the first DEHP reduction (Fig. 2). The overall concentration of PVC was kept low to minimize the viscosity effects (see Supplementary Table 15) and these data were measured in triplicate for multiple scan rates for each molar mass PVC (Supplementary Figs. 14–16). We observed that the reversibility of the first DEHP reduction decreased with increasing concentration of PVC, which suggests that the singly reduced DEHP (DEHP^{•-}) transfers an electron to the PVC (Fig. 2c) and, as a consequence, exhibits a lower return oxidation current. This trend was magnified at slower scan rates, wherein the DEHP^{•-} has more time to encounter PVC. Combined, these studies are consistent with DEHP serving as a redox mediator in the PVC reduction.

Bulk electrolysis shows plasticizer-mediated PVC reduction

Further evidence of DEHP's role as a redox mediator was obtained using a preparative-scale set-up that mimicked the electrosynthesis conditions. More specifically, we performed bulk electrolysis experiments of a divided cell that contained DEHP and/or PVC at the cathode/working electrode, and arene **1** at the anode/counter electrode (Fig. 2d). These experiments were used to determine the voltage required to generate a set current (galvanostatic conditions; Fig. 2e) and vice versa (potentiostatic conditions; Fig. 2f). Under constant-current conditions, DEHP alone generated -2.6 V of overpotential until reaching its theoretical capacity (6.7 mA h; see page 27 of the Supplementary Information). Gas chromatography–mass spectrometry (GC-MS) analysis after electrolysis showed that DEHP was partially degraded into 2-ethylhexanol and presumably other undetectable fragments, and that arene **1** remained unreacted at the anode (Supplementary Fig. 17). When PVC_{47k} (8 equiv.) was added to DEHP (1 equiv.) and electrolysed under the same current, a slightly smaller overpotential (-2.5 V) was observed and the capacity was substantially increased (24 mA h), indicating that the mixed system generated more electrons than DEHP alone. Post-reaction analysis

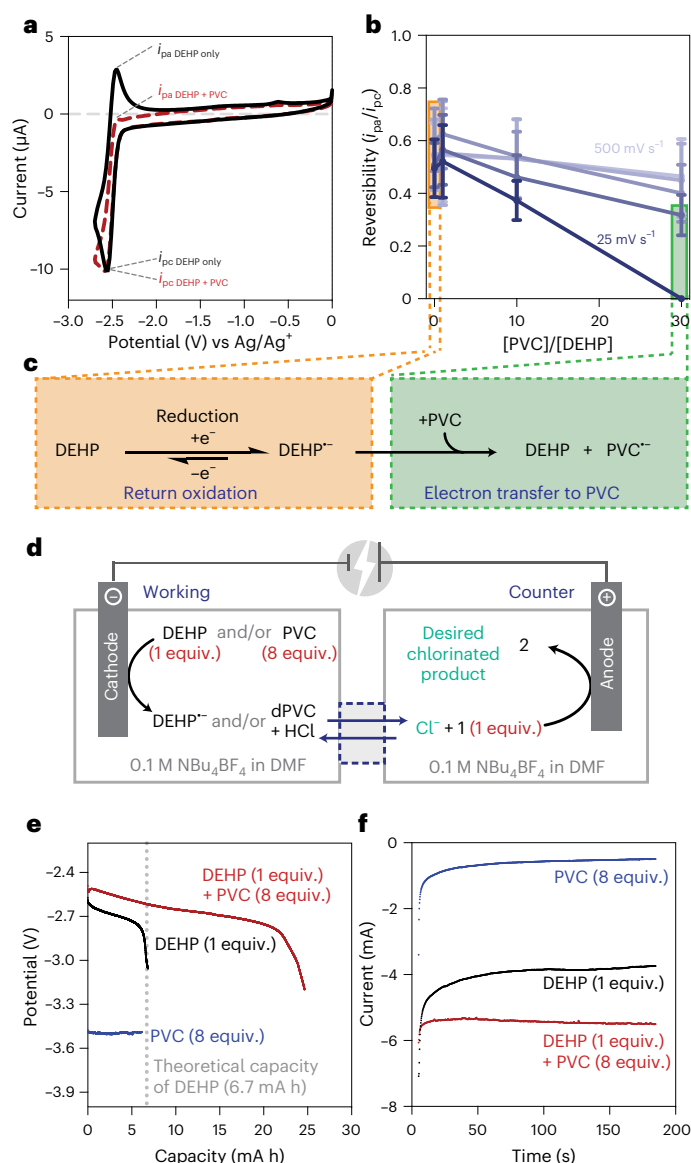


Fig. 2 | Electroanalytical data. a–c, Representative cyclic voltammetry data: cyclic voltammograms of the first reduction of DEHP with and without PVC_{47k} (30 equiv.) collected at 100 mV s⁻¹ (a), redox reversibility as a function of excess factor measured in triplicate (see Supplementary Table 17 for data with s.d.) at several scan rates (25, 50, 100, 250 and 500 mV s⁻¹) (b), and a scheme depicting the standard electrochemical pathway for the single reduction of DEHP (orange) and the alternative pathway when PVC is added (green), which is suggested by the cyclic voltammetry data (c). **d–f**, Bulk electrolysis data: schematic of the bulk electrolysis set-up and reactions occurring at each electrode (d), voltage readout from -5 mA constant-current electrolysis (e) and current readout from -2.5 V constant-voltage electrolysis (f). These data show that PVC can be reduced at smaller voltages with DEHP present, indicating that DEHP functions as a redox mediator.

revealed less DEHP degradation than without PVC, and quantitative conversion of arene **1** to chlorinated arene **2**. By contrast, bulk electrolysis on PVC alone required a much larger voltage (-3.5 V) to reach the same current. This voltage matches the solvent potential window, where solvent and/or electrolyte degradation probably occurs. Indeed, GC-MS analysis showed that tributylamine was formed at the cathode (from electrolyte degradation³⁸) and arene **1** remained unreacted at the anode. The inability to chlorinate arene **1** with PVC alone in the bulk electrolysis set-up differs from the galvanostatic synthetic trials, which proceeded with or without DEHP, albeit to different yields. It is

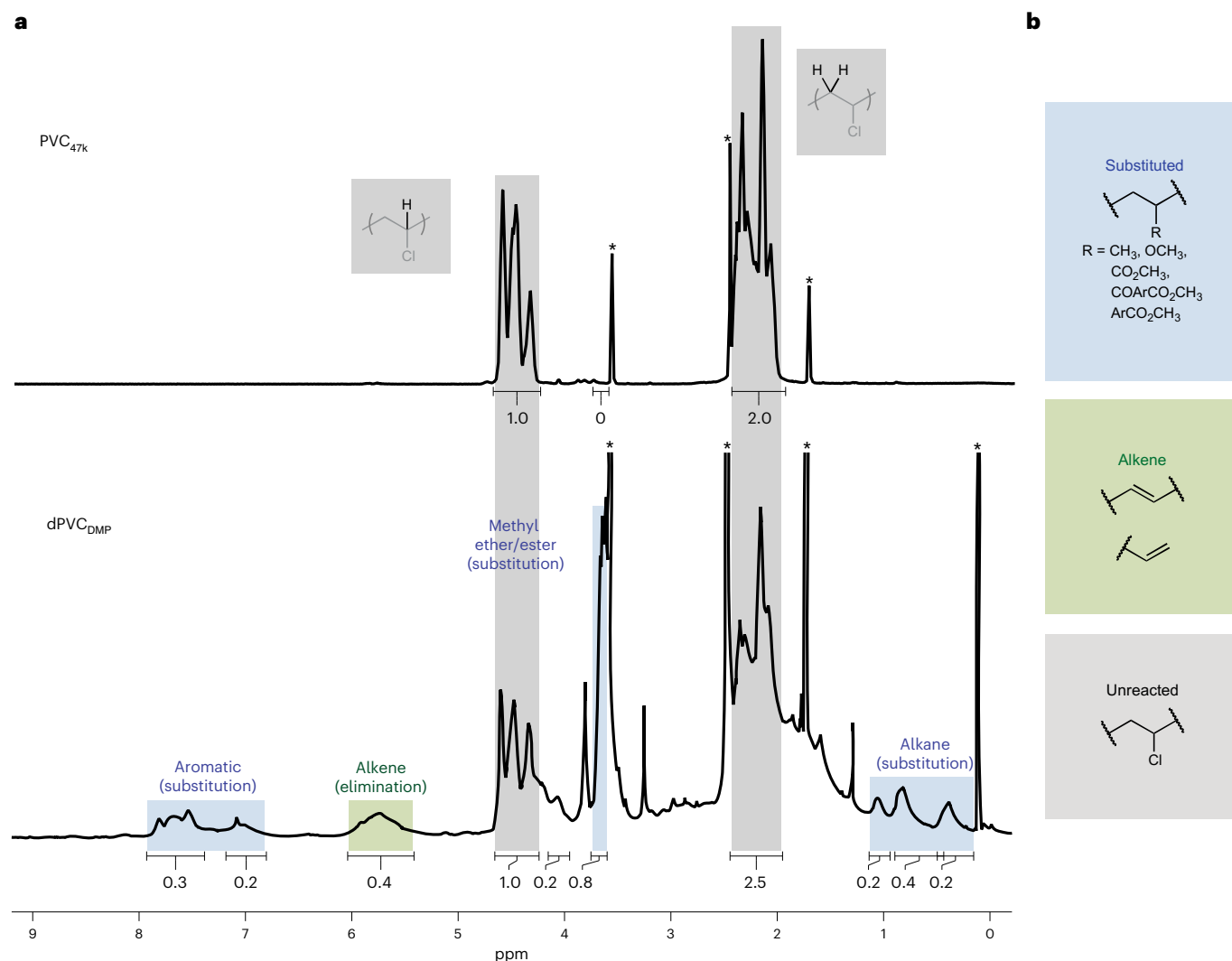


Fig. 3 | Structural information for PVC_{47k} and dPVC_{DMP}. **a**, ¹H NMR spectra (500 MHz, THF-*d*₆). **b**, The chemical structures of probable repeat units in dPVC_{DMP}—including a complex mixture of substituents as well as unreacted PVC units—compared with unreacted PVC_{47k}. Note that a large excess of phthalate was

used to increase the NMR signal intensity of substituted products in the resultant polymer (dPVC_{DMP}). The peaks associated with the solvent (THF), water and silicone grease are marked by asterisks.

possible that the exclusion of O₂ and H₂O in the bulk electrolysis set-up, but not the synthetic trials, may have affected PVC degradation and arene chlorination. Bulk electrolysis experiments were also performed under constant-voltage to simulate potentiostatic reaction conditions (Fig. 2f). Each electrolysis was conducted for several minutes until a steady-state current was reached. When the cathode was set to −2.5 V, DEHP alone generated −3.8 mA of reduction current. Meanwhile, PVC alone produced a much smaller reduction current (−0.5 mA) at this voltage, indicating essentially no electrochemical activity. The largest reduction current (−5.5 mA) was generated when DEHP was mixed with PVC, indicating that more electrons were flowing through the system. These data again support that the PVC—when restricted by a voltage cutoff—can only be reduced when DEHP is present.

Polymer characterization after electrolysis

After electrolysis, the resulting polymer (dechlorinated PVC, dPVC) was characterized to determine the extent of dechlorination and any new structural features. Briefly, the soluble polymer recovered from the standard galvanostatic reaction listed in Fig. 1—performed with DEHP (dPVC_{DEHP}) and without DEHP (dPVC_{alone})—was isolated and characterized. Neither dPVC sample showed a substantial change in *M_n* relative

to the unreacted PVC via size-exclusion chromatography (Supplementary Fig. 21), which is unsurprising given the large excess of PVC used and expected ~20% conversion of functional groups within each chain. Elemental analyses performed on the soluble residue indicated that both dPVC samples had reduced Cl content (dPVC_{DEHP}, 44% Cl; dPVC_{alone}, 53% Cl) relative to unreacted PVC (57% Cl) (Supplementary Table 20). Similarly, less mass (HCl) was lost at lower temperatures in the thermal gravimetric analysis for both dPVC samples compared with unreacted PVC, consistent with dechlorination (Supplementary Fig. 22). Differential scanning calorimetry (DSC) data showed a lower glass-transition temperature for dPVC_{DEHP} (*T_g* = 59 °C) relative to unreacted PVC (*T_g* = 83 °C), suggesting internal plasticization, potentially due to the 2-ethylhexyl side-chains (Supplementary Fig. 25)³⁹. Indeed, both ¹H NMR and infrared spectra of dPVC_{DEHP} showed evidence of 2-ethylhexyl substitution (Supplementary Fig. 23). One consequence is that these substituents will increase the carbon content and lead to an overestimation of the dechlorination in the elemental analysis. Both dPVC samples showed a much higher degree of crystallinity relative to unreacted PVC (Supplementary Fig. 25).

To further elucidate the polymer structure, a tenfold excess of dimethylphthalate (DMP) was used in the electrolysis reaction to generate

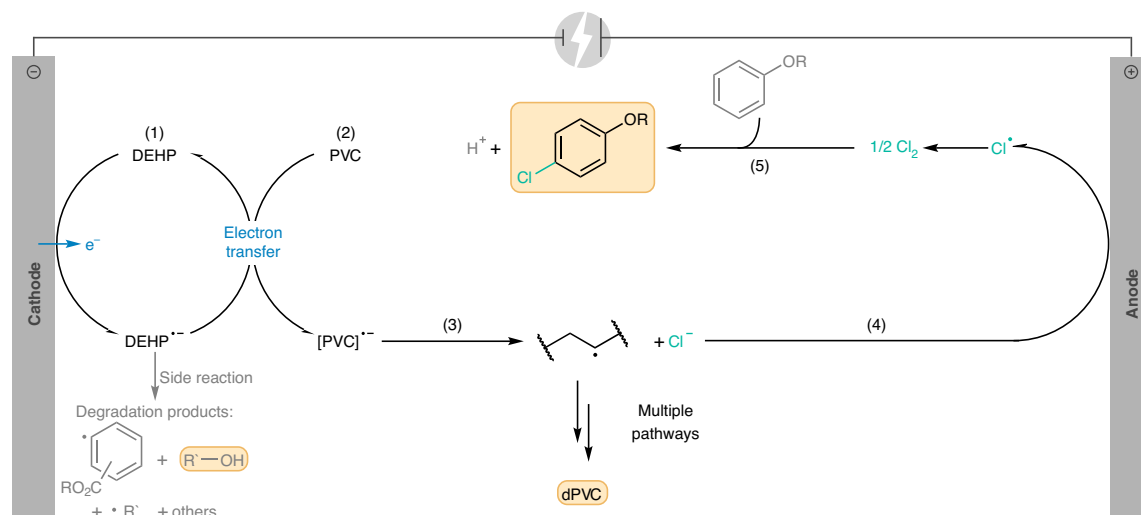


Fig. 4 | Proposed redox-mediated paired-electrolysis mechanism. Key steps are numbered: (1) reduction of plasticizer; (2) electron transfer to polymer; (3) dechlorination of polymer; (4) chloride oxidation; and (5) oxidative chlorination of the arene. Major products observed from the reaction are highlighted in the yellow boxes.

dPVC_{DMP}. The increased phthalate loading was intended to increase the signal intensity of substituted products and the methyl ester side-chain was intended to simplify the aliphatic region of the ¹H NMR spectrum. A substantial decrease in molar mass was observed via size-exclusion chromatography ($M_n = 14 \text{ kg mol}^{-1}$; Supplementary Fig. 26) compared with the unreacted PVC_{47k}, suggesting some chain scission was occurring. The ¹H NMR spectrum reveals a complex mixture of substituents, including alkenyl, aromatic, ester/ether and alkyl groups, alongside unreacted PVC units (Fig. 3). The possible mechanism(s) leading to these substitution patterns is discussed in more detail below.

Proposed mechanism

Taken together, the electrosynthesis, electroanalytical and characterization experiments suggest that two different pathways are operable for the initial PVC reduction. With DEHP present, an indirect reduction of PVC occurs wherein DEHP transfers the electron between the cathode and PVC (Fig. 4). Without DEHP present, PVC reduction requires a much larger voltage, and may react through direct interaction at the electrode, or with the solvent/electrolyte degradation products that were observed at these extreme voltages (Supplementary Fig. 17)^{31,38}. Ultimately, the milder and more controlled conditions offered by the galvanostatic DEHP-mediated pathway are preferred. Furthermore, this pathway effectively repurposes the plasticizer in plastic waste by using it as a redox mediator.

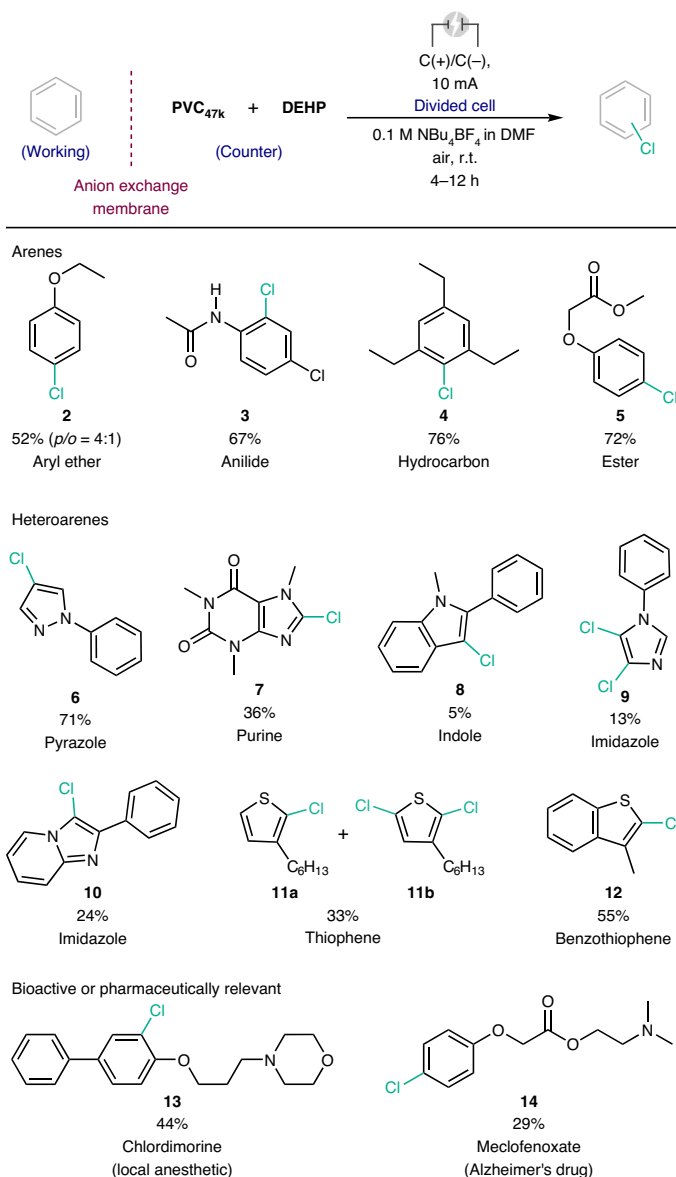
Regardless of how the electron is added to PVC, upon reduction, the chloride anion (Cl^-) is probably cleaved from the PVC via a concerted electron-transfer–bond-breaking process, as has been suggested³⁷ for small-molecule alkyl halides. The resulting carbon-based macroradical should be highly reactive and probably undergoes several reaction pathways (Supplementary Fig. 28). Based on NMR spectral data, the PVC macroradical reacts with alkyl and aryl radicals generated from the concomitant and unavoidable DEHP degradation³⁶. The PVC macroradical may also undergo scission reactions to form lower-molar-mass polymers, which in turn generate alkenes and end-group macroradicals that probably disproportionate to generate alkene and alkane end-groups. Poly(vinyl chloride)-based macroradicals may also react with O_2 or H_2O , and further facilitate dehydrochlorination, as has been observed in other systems⁴⁰. Depending on the location, the formed alkenes may generate allylic C–Cl bonds, which would be even more susceptible to reductive bond cleavage⁴¹. The observed acidification of the reaction medium suggests that H^+ is also released from the polymer upon reduction, perhaps through a concerted deprotonation/

oxidation (Supplementary Fig. 28). Notably, the net release of HCl from PVC is a common reaction pathway with nucleophiles, heat, base or acid⁴². Control electrolysis reactions wherein aqueous HCl is used in place of PVC proceed to high yields, further supporting the role of Cl^- ions (Supplementary Table 21).

The released Cl^- ions ultimately chlorinate arene substrates at the anode. Oxidation of Cl^- to either chlorine (Cl_2) or hypochlorite (OCl^-) intermediates have been proposed as the active chlorine species in electrochemical chlorinations^{23,43–46}. Under our open-to-air reaction conditions, any Cl_2 formed may react with adventitious water to form HOCl (ref. 47). Control reactions wherein Cl_2 or NaOCl was used as a chlorine source both proceeded without applied potential, further supporting that either could be the active chlorine species (Supplementary Table 21). When TEMPO was added, no product was formed in the PVC/DEHP reaction nor the HCl control reaction, suggesting that radical intermediates may be involved in the oxidative chlorination, either to convert Cl^- into a Cl radical or in the arene chlorination (Supplementary Table 21).

Substrate scope

Chlorinated arenes are found in pharmaceuticals, agrochemicals, disinfectants, dyes, electrical goods and solvents⁴⁸. To determine the scope of this method and probe possible interference from either PVC and/or DEHP, we screened various aromatic and heteroaromatic substrates (Table 1). In an undivided cell, some substrates were consumed during electrolysis but did not yield any chlorinated products with PVC/DEHP, but did when HCl was used (for example, anilide 3). These studies suggest that some substrates may react directly with PVC and/or DEHP under the reductive conditions. To avoid this unintended reactivity, all subsequent reactions were performed in a divided cell in which the working and counter compartments were separated by an anion-exchange membrane. Compound classes that have been employed in other oxidative chlorination reactions (for example, substituted benzenes, thiophenes and several classes of N-heterocycles) were successfully employed herein using PVC as a chloride source^{23,25,45}. Some compounds, including indole derivatives, generated lower yields because other oxidative reaction pathways (dimerization/oligomerization) competed with chlorination. Chlorinated pharmaceuticals²⁵ 13 and 14 could be directly formed under these conditions from their non-halogenated precursors. Although the yields need to be further optimized, the broad substrate scope and functional group tolerance of this method is promising.

Table 1 | Schematic of the divided-cell reaction set-up for the paired-electrolysis reaction and substrates investigated in this work

Isolated yields are listed.

Commercial PVC tubing as chloride source

To determine whether this approach can be used with PVC waste, we used PVC tubing (PVC_{tubing}) commonly used for liquid and gas transport in the research, medical, and food and beverage industries (Fig. 5a). The PVC_{tubing} used herein contained 32% liquid additives by mass, determined to be DEHP by ¹H and ¹³C NMR spectroscopic analysis (Supplementary Figs. 2–4). Once the plasticizer was removed, the resulting solid material (PVC_{solid}) had a *M_n* of 83 kg mol⁻¹ and was partially soluble in DMF. A 32% yield of **2** was observed when the galvanostatic reaction was performed using PVC_{solid} alone (Fig. 5b), similar to the 29% yield observed previously with PVC_{100k} alone. When PVC_{tubing} was directly used—with the same total mass of polymer—the yield of **2** substantially increased (to 63%). These results suggest that plasticized PVC waste can be used directly without needing to add more plasticizer. Although some manufacturers are moving towards phthalate-free formulations⁴⁹,

this legacy additive is still prevalent in most waste streams. Moreover, the dual-role of DEHP as plasticizer and redox mediator demonstrates that additives can serve one function during the lifetime of a plastic, and then be activated to serve a second function at the plastic's end-of-life. Moving forward, multipurpose plastic additives can be imagined that facilitate end-of-life treatment of plastics, which could considerably improve the overall efficiency of plastics reuse and repurposing.

Mixed plastic waste can be used

To simulate a mixed-plastic-waste stream, we evaluated the reaction efficiency with other plastic waste added (Fig. 5c). The overall yield of **2** was unchanged when similar masses of PET, HDPE, LDPE, LLDPE, PP and PS were added to the reaction mixture. Moreover, the plastic pieces made from PET, HDPE, LDPE, LLDPE and PP were insoluble in DMF and could be easily filtered out after the reaction was complete. These reactivity and solubility differences enable a facile separation of PVC from most other waste plastics.

LCA

A preliminary comparative LCA was conducted to estimate the potential environmental impacts of this approach. A simplified assessment was performed because this electrochemical process is still in the early development stage⁵⁰. More specifically, we sought to estimate the technology's decarbonization potential, which will help determine whether this waste diversion strategy should be pursued further. Three different scenarios for the electrosynthetic production of **2** were compared: scenario 1 uses HCl as the chloride source (3.2 equiv. Cl⁻ versus **2**); scenario 2 uses PVC waste as the chloride source (8 equiv. Cl⁻ versus **2**) and the leftover dPVC is landfilled; and scenario 3 reflects a hypothetical, ideal, atom-economical situation in which PVC waste (1 equiv. Cl⁻ versus **2**) is used as the chloride source with a 100% dechlorination efficiency and the residual hydrocarbon content (dPVC) is pyrolysed. Note that the hypothetically unused 7 equiv. of PVC is still landfilled in scenario 3. For all three scenarios, the functional unit was defined as 1 kg of chlorinated arene **2**. The impact analysis was performed by tracking the greenhouse gas emissions using the GWP metrics in kgCO₂e. The relevant data were gathered from the Ecoinvent libraries in the LCA software, SimaPro v9.1.1 (ref. ⁵¹), as well as the literature. More details on the scope, system boundaries, assumptions, life-cycle inventories and life-cycle impact assessments are presented in pages 47–49 of the Supplementary Information. It is important to note that this LCA only considered chloride source materials and end-of-life stages, under the assumption that the remaining stages and processes are constant for all scenarios.

With these caveats in mind, the LCA showed that using PVC waste as the chloride source (scenario 2) decreases the GWP by 71% compared with HCl (scenario 1) in the analysed stages. This difference arises largely because HCl production has a GWP that is over 11-times higher than the supply chain of PVC waste. In hypothetical scenario 3—in which PVC is fully dechlorinated and it is possible to pyrolyse the hydrocarbon content of the residual polymer—the kgCO₂e of **2** decreases by 56% compared with HCl. The reason for the reduced impact (versus scenario 2) is because mixed-plastics pyrolysis has a higher GWP than landfilling due to its process energy requirements. The results of scenario 3 point to improvements needed in pyrolysis technology alongside PVC dechlorination efficiency. Overall, these LCA results suggest that this electrochemical method is a promising approach for obtaining value from PVC waste. A more rigorous LCA will be performed in the future, alongside continued development of this process, including detailed characterization of all waste effluents. Future assessments with more environmental metrics could reveal benefits (for example, in human toxicity or freshwater ecotoxicity) from transitioning from landfilling to chemical recycling (repurposing) of PVC waste.

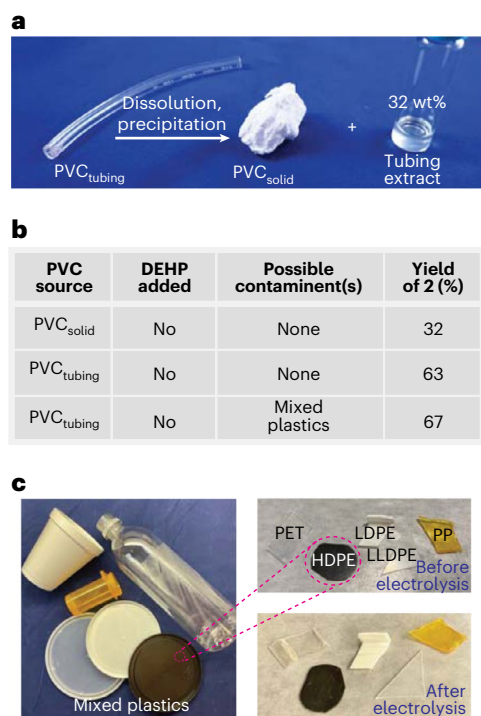


Fig. 5 | Evaluating real plastics in the paired-electrolysis reaction. a, Poly(vinyl chloride) tubing used to simulate PVC waste. **b**, Yields of galvanostatic reaction using arene **1** as the chlorination substrate. **c**, Plastic items used to simulate a mixed-plastics-waste stream. The mixed-plastics reaction contained by mass: 10% PET, 13% HDPE, 15% LDPE, 9% LLDPE, 13% polypropylene (PP), 9% polystyrene (not shown) and 31% PVC. The higher yield for PVC_{tubing} compared with PVC_{solid} demonstrates that plasticized PVC waste can be used directly without adding plasticizer.

Conclusions

Overall, this work demonstrates a proof-of-concept method for repurposing PVC waste wherein the PVC is used as a solid reagent in a paired-electrolysis reaction to produce value-added chlorinated organic products in a room temperature, air- and moisture-tolerant, single-pot process. These results also show that DEHP, which is currently used as a PVC plasticizer, functions in an advantageous role as a redox mediator to improve the synthetic yields. As a result, the plastic additive is effectively repurposed as well. Ultimately, the power of DEHP as a redox mediator is limited by its competitive degradation. Developing more stable redox mediators should enable lower mediator loadings and improved PVC dehydrochlorination yields. Efforts along these lines are underway.

A simplified LCA of only the materials and end-of-life stages revealed that the electrochlorination using PVC exhibited a 71% reduction in GWP (CO₂e released per kilogram of chlorinated product) compared with the analogous process using pristine HCl. Moreover, with further optimization to generate more alkenes and fewer substituted products, the partially or fully dehydrochlorinated PVC may also be a valuable resource, potentially as a reactive polymer for hydrogenation (to polyethylene), cross-linking (elastomers), metathesis (depolymerization) or other functionalization reactions. Most importantly, the new functionality of DEHP uncovered in this work motivates a rethinking of plastic additives in chemical recycling, wherein additives are intentionally designed to facilitate a future recycling process.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information,

acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41557-022-01078-w>.

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Methods

Electrosynthesis

Bulk electrolysis was performed using an ElectraSyn 2.0 and electrodes supplied by IKA. Reactions performed under N_2 were first degassed by bubbling N_2 for 30 min, and then electrolysis was performed under a N_2 -filled balloon. All other reactions were performed opened to air. Anhydrous DMF (Sigma-Aldrich, lot no. SHBL1229) was stored in a sure/seal bottle and used in reactions performed under N_2 . American Chemical Society reagent-grade DMF (Sigma-Aldrich, lot no. SHBD7622V) was used as received in reactions performed open to air. NBu_4BF_4 (Oakwood Chemical, lot no. 007316L07W) was used as electrolyte in all reactions. Undivided cell reactions were performed using a 10 ml IKA ElectraSyn vial. Divided-cell reactions were performed using an IKA Pro-Divide and a 5 μ m frit. A FuelCell-Store PiperION anion-exchange membrane, 15 μ m, mechanically reinforced, was pre-soaked in 0.1 M NBu_4BF_4 overnight and layered with the frit. A detailed assembly guide is provided in the Supplementary Information.

Conditions for an undivided cell paired-electrolysis, constant-current, 8 ml scale: PVC (200 mg, 3.2 mmol (repeat unit), 8.1 equiv. (repeat unit)), NBu_4BF_4 (264 mg, 0.80 mmol), DEHP (0.157 ml, 0.4 mmol, 1.0 equiv.), ethoxybenzene (0.051 ml, 0.4 mmol), tridecane internal standard (0.050 ml) and DMF (8 ml) were added to a 10 ml ElectraSyn reaction vessel equipped with Teflon coated magnetic stir bar (egg-shaped, 16 mm length) and two graphite electrodes ($0.8 \times 0.2 \times 5.20$ cm). All reagents were dissolved by stirring for at least 15 min, then the reaction mixture was subjected to constant-current electrolysis (10 mA) with alternating polarity (15 min) at room temperature. The reaction was stirred at 400 r.p.m. for 16 h.

Conditions for an undivided cell paired-electrolysis, constant-voltage, 8 ml scale: PVC (200 mg, 3.2 mmol (repeat unit), 8.1 equiv. (repeat unit)), NBu_4BF_4 (264 g, 0.80 mmol), DEHP (0.157 ml, 0.4 mmol, 1.0 equiv.), ethoxybenzene (0.4 mmol), tridecane internal standard (0.050 ml) and DMF (8 ml) were added to a 10 ml ElectraSyn reaction vessel equipped with Teflon coated magnetic stir bar (egg-shaped, 16 mm length), two graphite electrodes ($0.8 \times 0.2 \times 5.20$ cm) and a silver wire reference electrode stored in 3 M KCl in water. All reagents were dissolved (or swelled) by stirring for at least 15 min, then the reaction mixture was subjected to constant-voltage electrolysis (-1.3 V) with alternating polarity (15 min) at room temperature. The reaction was stirred at 400 r.p.m. for 16 h.

Conditions for a divided cell paired-electrolysis, constant-current: the ElectraSyn Pro-Divide reaction vessel was assembled using graphite electrodes ($0.8 \times 0.2 \times 5.20$ cm) and a 5 μ m frit separator layered with a PiperION anion-exchange membrane (pre-soaked in 0.1 M NBu_4BF_4 in DMF). The aromatic substrate (0.5 mmol), NBu_4BF_4 (231 mg, 0.70 mmol), and DMF (7 ml) were added to the working (anodic) chamber; PVC_{47k} (200 mg, 3.2 mmol (repeat unit), 6.4 equiv. (repeat unit)), NBu_4BF_4 (231 g, 0.70 mmol), DEHP (0.20 ml, 0.50 mmol, 1.0 equiv.) and DMF (7 ml) were added to the counter (cathodic) chamber. All reagents were added under air and then dissolved by stirring for at least 15 min. Then the reaction mixture was subjected to constant-current electrolysis (10 mA) at room temperature. The anodic reaction was monitored by gas chromatography and stirred until all the aromatic substrate was consumed. The working (anodic) chamber was poured into -250 ml of ethyl acetate and washed with H_2O (5×40 ml) and brine (1×50 ml) to remove DMF. The organic layer was dried with $MgSO_4$, filtered, and concentrated under reduced pressure. The crude reaction was further purified by column chromatography and the final product dried under reduced pressure.

Cyclic voltammetry

Dimethylformamide, DMF (99.8%, anhydrous) was obtained from Sigma-Aldrich and used as received. Tetrabutylammonium

tetrafluoroborate, NBu_4BF_4 (electrochemical grade) was obtained from Sigma-Aldrich and was dried under high vacuum for 48 h before being transferred to a nitrogen-filled glovebox. A 0.1 M stock solution of the supporting electrolyte in DMF was prepared in a nitrogen-filled glovebox (Mbraun Labmaster, water <4 ppm and oxygen <0.5 ppm) and was dried over 3 Å molecular sieves for at least two days before use.

Cyclic voltammetry was performed in a nitrogen-filled glovebox using a Biologic VSP multichannel potentiostat/galvanostat using a three-electrode electrochemical cell, consisting of a glassy carbon disk working electrode (3 mm diameter, 0.07 cm² area, BASi), an Ag/Ag⁺ quasi-reference electrode (BASi) with 0.01 M AgBF₄ (Sigma-Aldrich) in dimethylformamide calibrated with Fc/Fc⁺, and a platinum wire counter electrode (BASi). All experiments were conducted in a 0.1 M NBu_4BF_4 stock electrolyte solution

Bulk electrolysis

Bulk electrolysis (charging) was performed in a H-cell, with VitraPore frit (10 mm $\times$ 2.8 mm $\times$ 0.7 mm, porosity 5). The working and counter electrodes were reticulated vitreous carbon 5.1 cm long (McMaster 100 PPI $0.64 \times 15.2 \times 15.2$ cm) and the reference electrode (Ag/Ag⁺) was used in the working side. The cycling current was set to -5 mA under a voltaic cutoff (-3.2 V or -4.0 V) and both chambers stirred at 700 r.p.m. during charging. The working chamber was loaded with active species (PVC_{47k} 0.125 g (8 equiv.) or DEHP 99 μ l (1 equiv.) or PVC_{47k} 0.125 g (8 equiv.) + DEHP 99 μ l (1 equiv.)) in 0.1 M NBu_4BF_4 in DMF (5 ml). The counter chamber counter chamber was loaded with ethoxybenzene 29 μ l (1 equiv.) in 0.1 M NBu_4BF_4 in DMF (5 ml).

Data availability

A detailed description of the materials, equipment and methods are provided in the Supplementary Information. Furthermore, the Supplementary Information contains full characterization data for small molecules and polymers, electrochemical data, reaction screening results, LCA results and all of the other data collected in this work. When possible, all of the original data are included. In some cases, several hundred raw data files were generated for a single plot, and therefore these results are instead summarized in the tables and figures within the manuscript and its Supplementary Information. Due to the large number of raw data files, they are available from the corresponding author on reasonable request.

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Author contributions

D.E.F. and A.J.M. conceptualized the research. D.E.F. performed all electrosynthetic experiments, data analysis, created the figures and wrote the original draft. D.K. performed all electroanalytical experiments, contributed to figure creation, data analysis and editing of the paper. The LCA was performed by S.I.C. and J.F.A. A.J.M. also contributed to the funding acquisition, data analysis, supervision and editing of the manuscript. The funders had no role in study design, data collection and analysis, the decision to publish nor the preparation of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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